AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 4 · TEACHER GUIDE

Trace Tables

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach each loop update as inspectable evidence.

Supply complete correct/faulty scripts and distinguish changing-turn sums from fixed additions before independent transfer.

Keep one habit: Look for initialization, excluded stops, actual output events and revealing iteration counts.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: **Trace variable** values before and after each **loop iteration**

Teach for **loop**s with finite integer ranges before adding more **loop** forms. Keep range values, body executions and **output** events separate. Reference: https://docs.python.org/3/tutorial/controlflow.html#the-range-function.

The examples use a positive default step and do not require lists or while **loop**s.

By the end, I can make a **trace table** for a for-**loop**, including zero **iteration**s.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/4, ai-2/5, python-bridge/6. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Python 3 editor and **loop**s.py
- Counters or numbered paper cards
- Blank **trace table** with turn total before and total after columns

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.

> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

Look for: **Trace** the selected branch in an if elif else chain

Reference example, not the only acceptable wording: if tests negative first; elif checks <= 12 only after that fails; else handles the rest.

Unindented print runs after any selected branch, whose assignment gives label a value.

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

Look for: Translate a short paper calculation into a Python script

Reference example, not the only acceptable wording: packs = 4; per_pack = 6; total = packs * per_pack; print(total), each on its own line. The first two bind quantities, the third computes 24 and the fourth displays 24.

Assignment alone produces no displayed **output** in this script.

It reads the old 2, adds 3 and stores 5. Assignment updates a value; it is not an algebraic equality.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Branching Logic

Precision in Algorithms

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation.
A student evaluates the condition using the initial value. Locate the divergence.

Concept to explain

A **trace table** records changing **state** and reveals reasoning errors.

Term	Meaning
variable	A named reference to a value that can be used or changed in a program.
state	The current values and information that determine what a process does next.
trace	A record of the successive states, decisions or information transformations in a process.
iteration	One repeat of a process, often followed by inspection and revision.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach loop state and output events with complete traces

> Keep changing-turn accumulation, the fixed-size jar update and the independent counter example distinct.

State the task and range semantics

Imagine adding numbered counters: first 1, then 2, then 3. A **loop** can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script.

Tracing repeated calculations prepares you to inspect the data **process**ing used in larger AI applications.

In for turn in range(1, 4), Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body.

This lesson uses integer range values with the default positive step of 1.

Read the complete accumulator script

```
total = 0
for turn in range(1, 4):
    total = total + turn
print(total)
```

Trace initialization and excluded stops

The **output** is 6. Before the first **iteration**, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the **loop**.

The table records **state**; it does not create extra instructions.

Range expression	Values supplied	Number of iterations
range(1, 4)	1, 2, 3	3
range(1, 2)	1	1
range(1, 1)	No values	0
range(4)	0, 1, 2, 3	4

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0.

Do not rely on a **variable** first assigned inside a **loop**: a zero-**iteration** run might never assign it.

> **Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?**

AI ACADEMY / AI-2/4

Loop totals after each iteration

	turn	total after update
First	0	4
Second	1	7
Third	2	10
Fourth	3	13

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Introduce the separate token jar

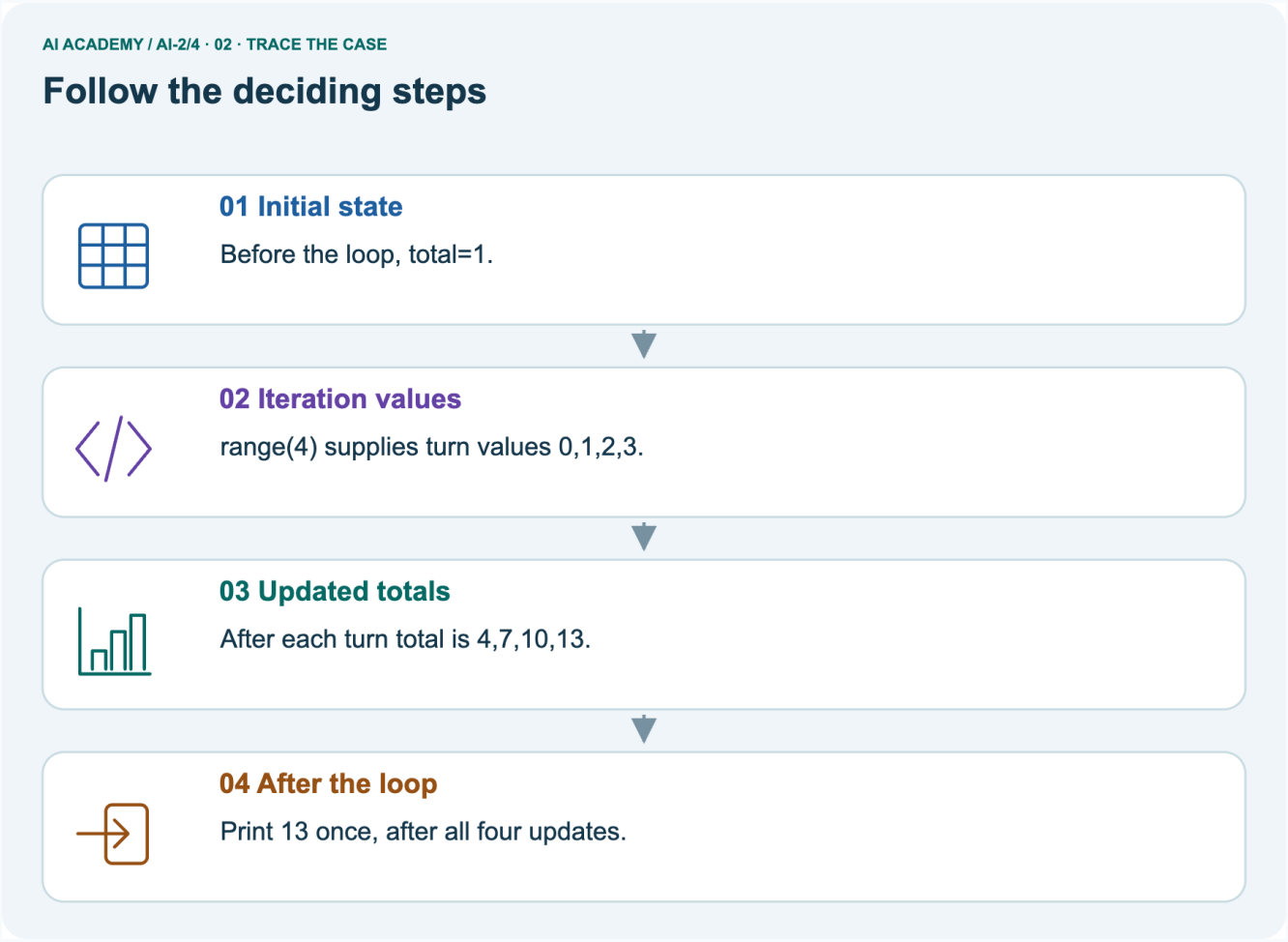
A fictional reading-token jar starts with one token and gains three per round.

Python: total=1; for turn in range(4): total=total+3; after the **loop** print(total).

The update is inside the **loop**; print is outside.

```
total = 1
for turn in range(4):
    total = total + 3
print(total)
```

Trace fixed additions and print placement



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Initial state	Before the loop, total=1.
2. Iteration values	range(4) supplies turn values 0,1,2,3.
3. Updated totals	After each turn total is 4,7,10,13.
4. After the loop	Print 13 once, after all four updates.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

The turn value labels an **iteration**; it is not the number added. The fixed update adds three each time. Four **iteration**s take the total from 1 to 13.

Keeping print outside the **loop** produces one final display rather than four intermediate displays. Initialise total before the **loop** so a zero-**iteration** run still has a defined value.

Limit claims to the stated finite loop

This finite **loop** does not establish behavior of a while-**loop** with a different update or stopping condition.

Use zero iterations to check initialization

Change: Use range(0) instead of range(4).

Prediction: The **loop** body never runs and the printed total is 1.

Reason: The iterable contains no turn values; the initial assignment still happened.

Connect the picture to actual state changes

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

2. Correct answer: score = score + 3 → 5.

3. Hint used: score = score – 1 → 4.

4. A table shows exactly where an unexpected value first appears.

Recalculate, re**trace**, or re**state** the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Set score=2; add 3; if score>4 subtract 1; double score.

Trace every value.

Exact explanation to model

2,5,4,8. The comparison is evaluated at 5 before subtraction.

Explain the added worked example

Key terms

Term	Meaning
Loop	A block of instructions that can run repeatedly
Iteration	One execution of a loop body
Accumulator	A variable that keeps a running result
Initialization	Giving a variable its starting value
Trace table	A record of variable values as instructions execute

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Trace Tables

The facts we will use

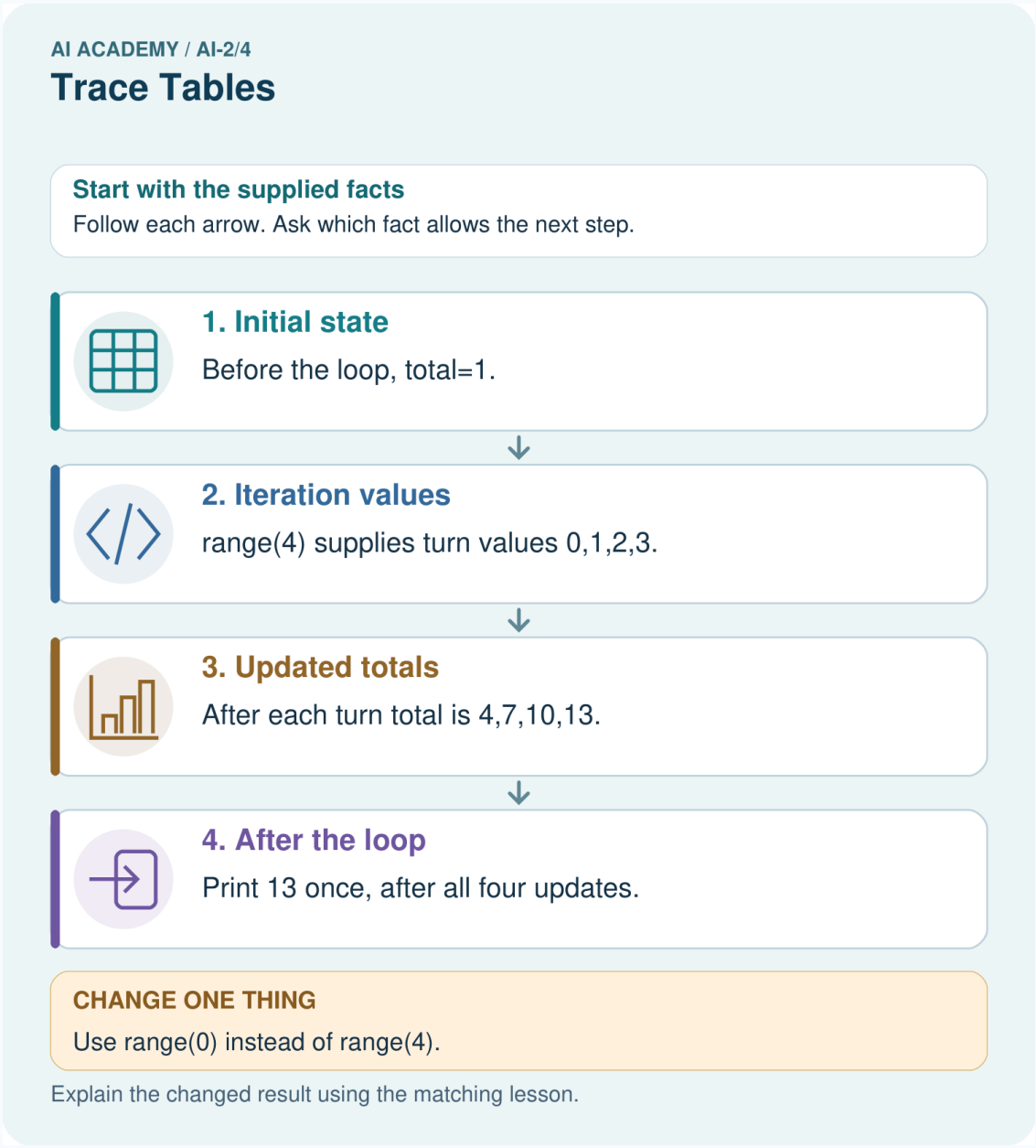
AI ACADEMY / AI-2/4

Loop totals after each iteration

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Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide the reasoning, then remove help

> Collect a response from each learner before discussing the worked key.

Prompt before revealing

Completion task: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Expected reasoning and fading help

Teacher check: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full **trace** only when the partial **trace** is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Check the misconception before discussion

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: List the supplied values before tracing; range(1, 4) excludes 4.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Use the pictorial check

Move print inside the original **loop**. What sequence is displayed?

4,7,10,13 on separate lines, because each print follows one update.

Changed-case reasoning: The **loop** body never runs and the printed total is 1. The iterable contains no turn values; the initial assignment still happened.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.
- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.
- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Set score=2; add 3; if score>4 subtract 1; double score.

Write the question to explain: **Trace** every value.

Reveal after **prediction**s: 2,5,4,8. The comparison is evaluated at 5 before subtraction.

Use the earlier changed question as optional later practice. Reserve Build a repeated addition independently for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, **trace** conditions for decisions, and identify supporting sources for claims.

Model these guided questions

range(1, 3) supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one **iteration** because there are no earlier additions to lose. range(1, 4) should give 6; the faulty version gives 3.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve assignment using the prerequisite.
7 to 20 minutes	Model range with numbered cards and complete W1.
20 to 33 minutes	Trace each row of the worked addition and predict output.
33 to 45 minutes	Complete W2 and compare predicted and observed values.
Session 2 start to 8 minutes	Retrieve included start and excluded stop.
8 to 20 minutes	Trace the reset bug, complete W3 and discuss the guided response.
20 to 35 minutes	Collect independent W4 code and zero one and several iteration traces.
35 to 45 minutes	Use W5 and the exit to examine assumptions and revealing tests.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Paper score trace is 2→5→4→8; evaluate >4 at 5. Correct main-loop outputs are 6,1,0 for range(1,4), range(1,2), range(1,1). The reset-bug version gives 3,1,0: one iteration hides the defect. Printing inside the corrected main loop shows 1,3,6; outside it shows only 6. The separate jar starts at 1 and adds 3 four times, giving 4,7,10,13; range(0) preserves 1. Preserve actual runs and make the source version clear. Reserve Build a repeated addition independently for the five-counter/two-per-round assessment.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Use columns for step, event, old score, new score, and branch. **Trace** once by hand, predict the next row, then compare.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim

Record	What to capture
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The stop value is included: List the supplied values before tracing; range(1, 4) excludes 4.

Initialization belongs inside every **iteration**: Resetting the **accumulator** loses earlier work.

An unindented print repeats with the body: Indentation determines which **state**ments repeat.

Zero **iteration**s mean the entire program does nothing: **Initialization** and **state**ments after the **loop** still execute.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.

What the response shows	Next teaching action
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Build a repeated addition independently for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a **trace**. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Initialize total = 5 before the **loop**; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final **output**s for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible **trace**.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the range

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

2, 3 and 4; the stop value is excluded. The body runs three times.

W2 Trace before running

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

W3 Repair the reset

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final **output** and remove only the faulty reset.

Faulty after-addition values are 1, 2, 3 because each starts from zero. Remove the indented total = 0. Correct after-addition values are 1, 3, 6; final **output** 6.

W5 Check a contract limitation

In your counter script, range(-2) performs no **iteration**s. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

No. The written contract requires a nonnegative integer. The script assumes this and does not reject negative values. Empty range behavior does not validate the **input**; report the limitation or add an explicit check.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

2,5,4,8. The comparison is evaluated at 5 before subtraction.

Additional transfer practice question

Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

Reasoned practice answer

The **trace** is 2, 5, 10. Doubling first gives 2, 4, 7. Operation order changes the result.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A **trace table** records only the final answer.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

Trace is 1,5,3,6. The condition sees 5, so subtraction runs. Using the initial 1 incorrectly skips subtraction and gives 10. A **variable**'s current **state**, not its first value, determines each operation.

Explanations for the additional workbook examples

W1 Explain the range

W2 Trace before running

W3 Repair the reset

W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Initialize total = 5 before the **loop**; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final **output**s for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

W5 Check a contract limitation

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Trace each book's genre, length, difficulty, and total score before sorting.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Trace recommender **loop**s with current item, eligibility, score and current best so each update can be explained.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank **trace** grid and permit oral explanation.

Extension: put print inside the correct **loop** and explain why zero rounds produces no **output** while an outside print still does.

W4 must preserve the initial 5, update by 2, **trace** three **iteration**s and print once. Require **output**s 5, 7, 11 for zero one and three rounds.

After support, retry with an initial 4 and additions of 3 for zero one and two rounds: **output**s 4, 7, 10. Record assistance before judging independence.

Plan the delayed check

Suggested return points in this level: ai-2/5, ai-2/7, ai-2/10. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 5: **Loop**s and Stopping. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 5 examines loop progress and stopping; Week 7 develops structured records and table schemas.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: **Trace variable** values before and after each **loop iteration**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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Practice and assessment reminders

Before class, solve the supported practice: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Reasoned practice answer: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Check every learner before discussion: Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: List the supplied values before tracing; range(1, 4) excludes 4.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Use range(0) instead of range(4). Require a **prediction** and a reason before comparing. Expected result: The **loop** body never runs and the printed total is 1.

Reason: The iterable contains no turn values; the initial assignment still happened.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, **trace**/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

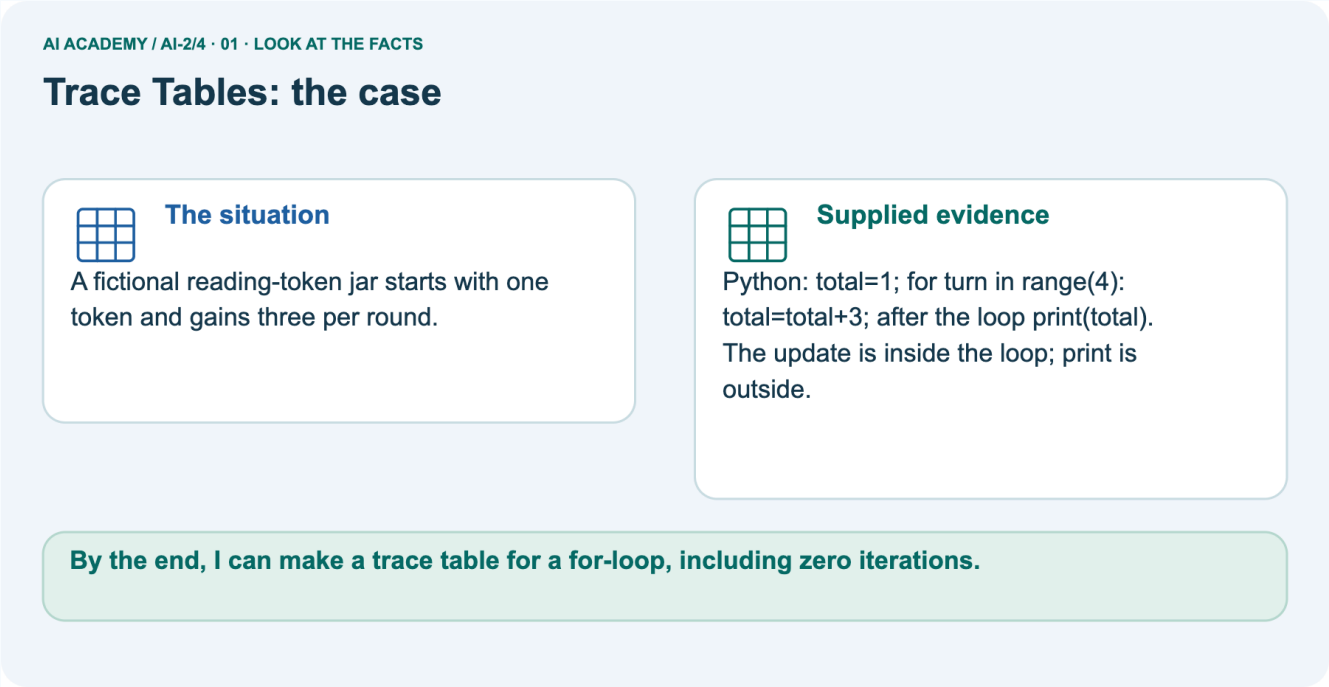
<details>
<summary>Reference: teach with the six pictures</summary>
</details>

REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Trace Tables: the case



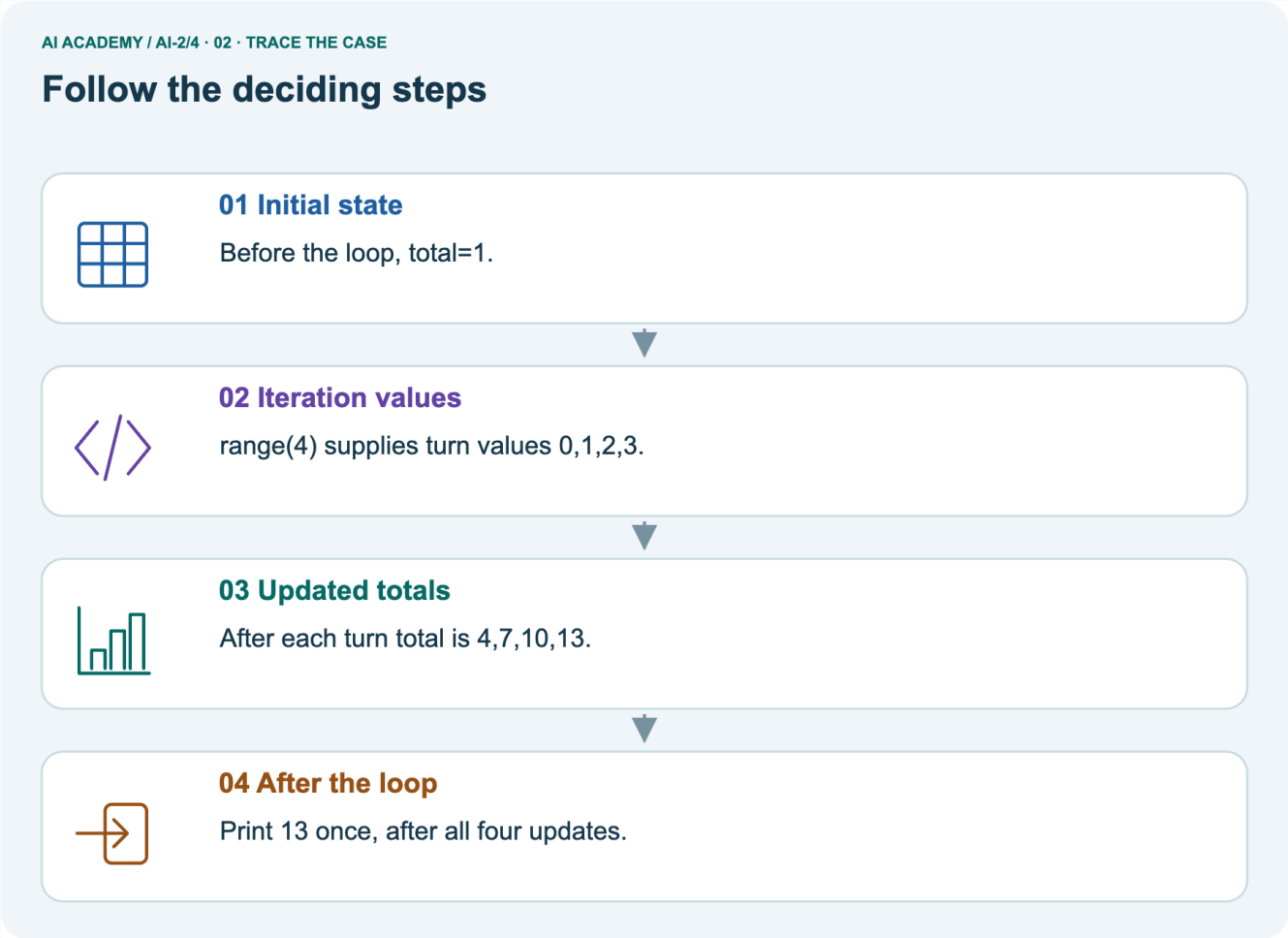
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional reading-token jar starts with one token and gains three per round. Supplied facts: Python: total=1; for turn in range(4): total=total+3; after the **loop** print(total). The update is inside the **loop**; print is outside.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



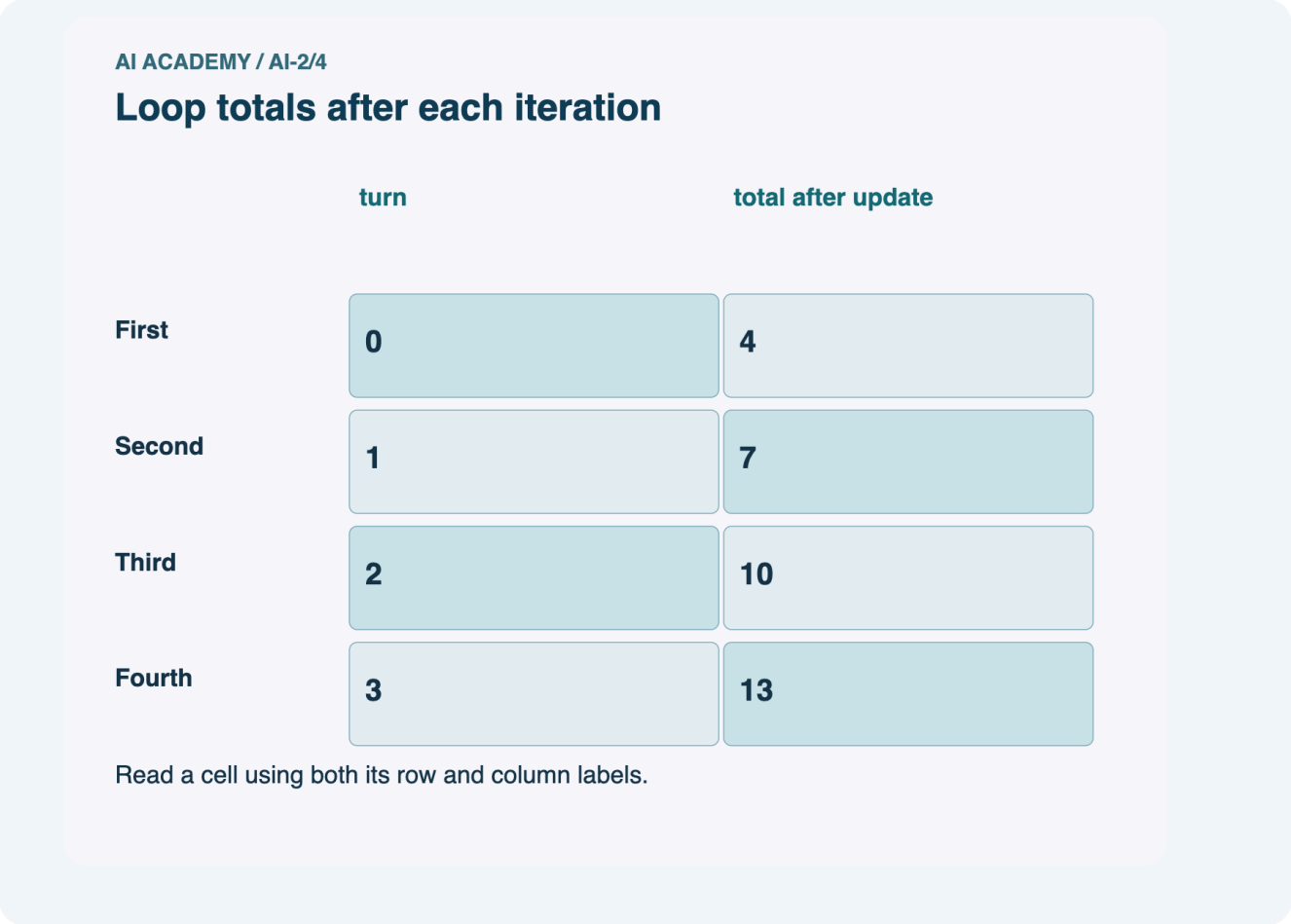
Follow the labelled picture with the matching explanation. Open the vector diagram

Initial **state**: Before the **loop**, total=1. → **Iteration** values: range(4) supplies turn values 0,1,2,3. → Updated totals: After each turn total is 4,7,10,13. → After the **loop**: Print 13 once, after all four updates.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Loop totals after each iteration



Follow the labelled picture with the matching explanation. Open the vector diagram

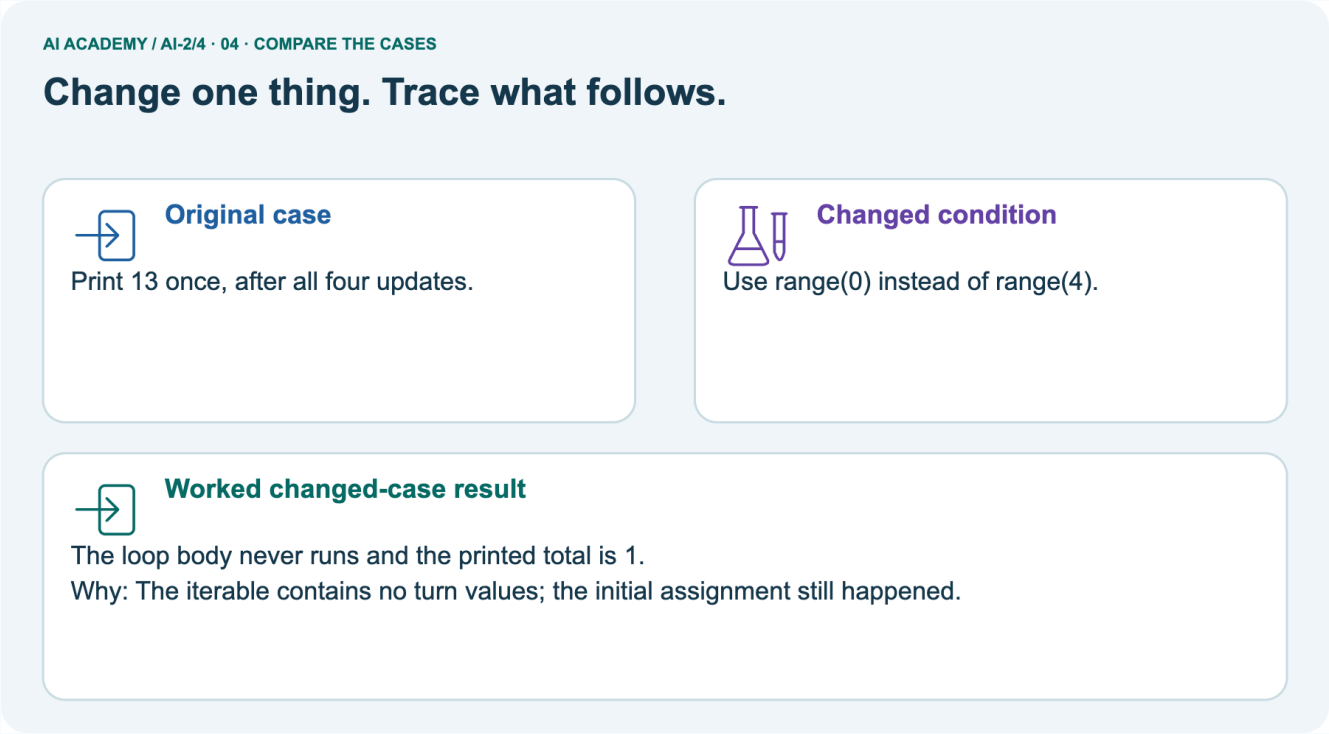
Loop totals after each **iteration**. First: turn = 0, total after update = 4. Second: turn = 1, total after update = 7. Third: turn = 2, total after update = 10. Fourth: turn = 3, total after update = 13.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Print 13 once, after all four updates. Changed condition: Use range(0) instead of range(4). Expected result: The **loop** body never runs and the printed total is 1.

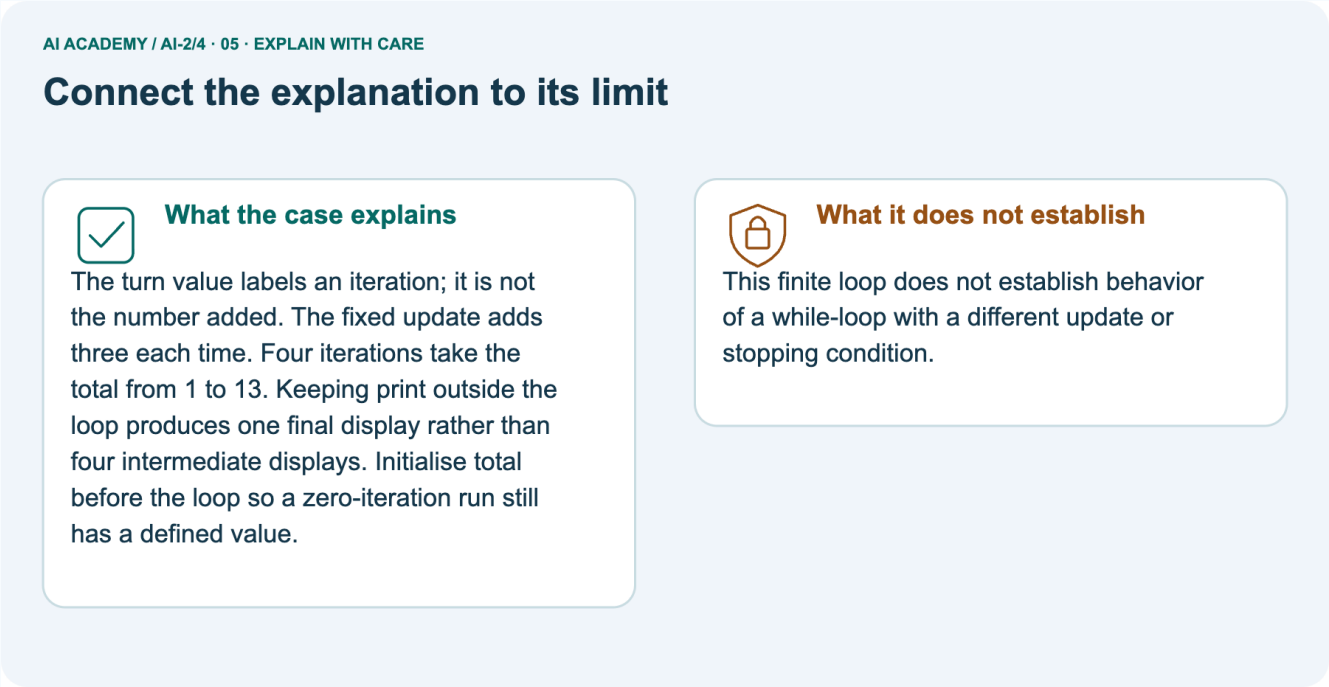
Reason: The iterable contains no turn values; the initial assignment still happened.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: The iterable contains no turn values; the initial assignment still happened.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The turn value labels an **iteration**; it is not the number added. The fixed update adds three each time. Four **iteration**s take the total from 1 to 13.

Keeping print outside the **loop** produces one final display rather than four intermediate displays. Initialise total before the **loop** so a zero-**iteration** run still has a defined value.

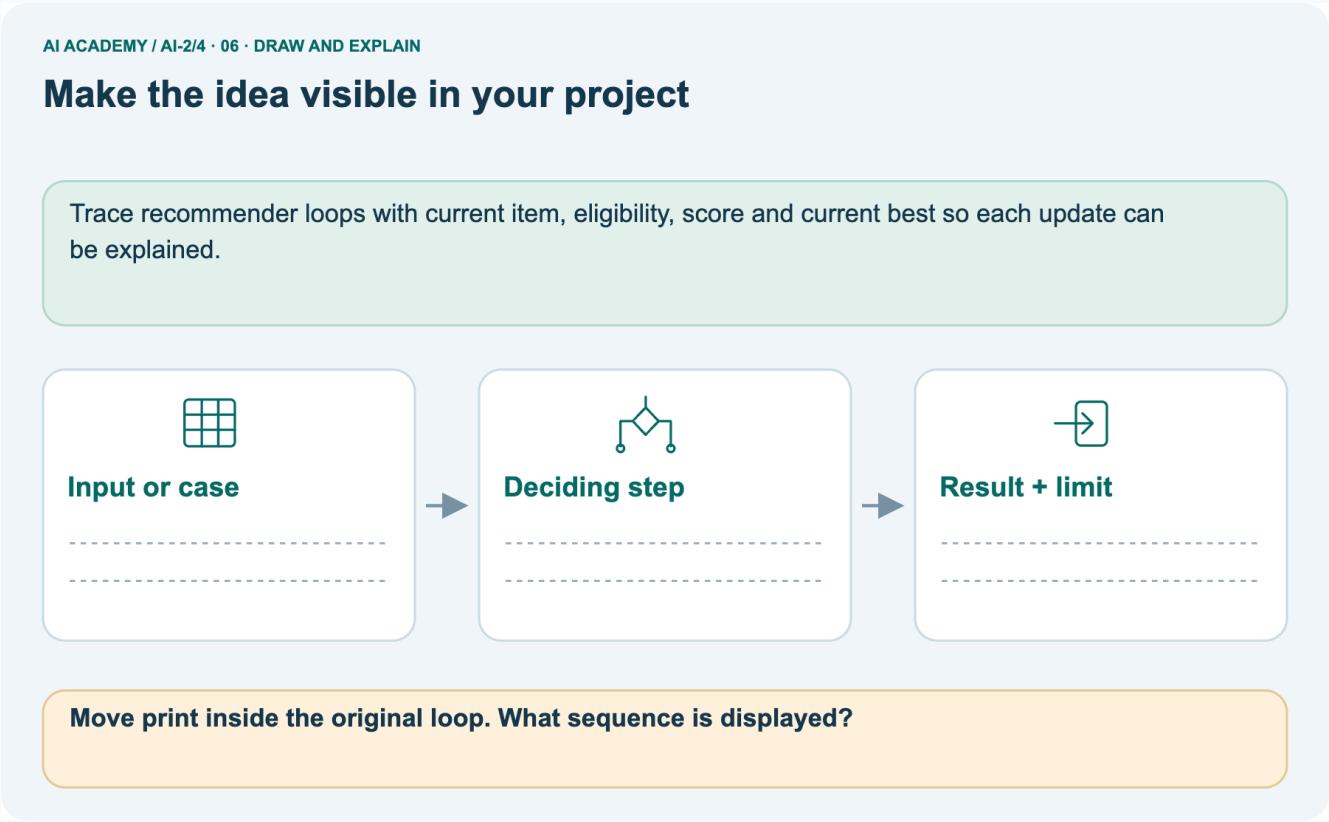
Limit: This finite **loop** does not establish behavior of a while-**loop** with a different update or stopping condition.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the **state**d scope: This finite **loop** does not establish behavior of a while-**loop** with a different update or stopping condition.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Trace recommender **loop**s with current item, eligibility, score and current best so each update can be explained. Draw your **input**, deciding step, result and limit. Supported question: Move print inside the original **loop**.

What sequence is displayed?

Draw your project **trace** and justify the deciding step using **evidence**.

Teacher prompt: Draw your project **trace** and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: 4,7,10,13 on separate lines, because each print follows one update.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Trace Tables: the case

02. Follow the deciding steps

03. Loop totals after each iteration

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.